Variation Method & FEA Course
Homework 1

Submitted before April 27, 2026

1. Using Lagrange formula $(\boldsymbol{a}\times\boldsymbol{b})\cdot(\boldsymbol{c}\times\boldsymbol{d})=(\boldsymbol{a}\cdot\boldsymbol{c})(\boldsymbol{b}\cdot\boldsymbol{d})-(\boldsymbol{a}\cdot\boldsymbol{d})(\boldsymbol{b}\cdot\boldsymbol{c})$ ($\boldsymbol{a},\boldsymbol{b},\boldsymbol{c}$ and $\boldsymbol{d}$ are all vectors), prove the following equation: $\boldsymbol{e_{ksp}e_{ipj}}=\boldsymbol{\delta_{is}\delta_{jk}}-\boldsymbol{\delta_{ik}\delta_{js}}$.

2. Let u_i be the displacement vector field. Prove that the infinitesimal strain

$$\varepsilon_{ij}=\frac{1}{2}(u_{i,j}+u_{j,i})$$

is a second-order tensor.

3. Prove that the permutation symbol $\boldsymbol{e_{ijk}}$ is a three-order tensor. Note that:

$$\boldsymbol{e_{ijk}}=\begin{cases}+1, & if\ (i,j,k)\ is\ an\ even\ permutation\ of\ (1,2,3)\\ -1, & if\ (i,j,k)\ is\ an\ odd\ permutation\ of\ (1,2,3)\\ 0, & otherwise\end{cases}$$

4. For a given tensor $T=\begin{bmatrix}T_{11} & T_{12} & T_{13}\\ T_{21} & T_{22} & T_{23}\\ T_{31} & T_{32} & T_{33}\end{bmatrix}=\begin{bmatrix}4 & -1 & 2\\ -1 & 3 & 0\\ 2 & 0 & 5\end{bmatrix}$, find the values of the following expressions:

(1) T_{ii};

(2) $T_{ij}T_{ij}$;

(3) $T_{ij}T_{ji}$.